



Numerical assessment of tsunami attack on a rubble mound breakwater using OpenFOAM[®]

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ABSTRACT

A numerical assessment study of tsunami attack on the rubble mound breakwater of Haydarpaşa Port, located at the southern entrance of the Istanbul Bosphorus Strait in the Sea of Marmara, Turkey, is carried out in this study using a Volume-Averaged Reynolds-Averaged Navier-Stokes solver, IHFOAM, developed in OpenFOAM[®] environment. The numerical model is calibrated with and validated against the data from solitary wave and tsunami overflow experiments representing tsunami attack. Furthermore, attack of a potential tsunami near Haydarpaşa Port is simulated to investigate effects of a more realistic tsunami that cannot be generated in a wave flume with the present state of the art technology. Discussions on practical engineering applications of this type of numerical modeling studies are given focusing on pressure distributions around the crown-wall of the rubble mound breakwater, and the forces acting on the single stone located behind the crown-wall at the rear side of the breakwater. Numerical modeling of stability/failure mechanism of the overall cross-section is studied throughout the paper.

The present study shows that hydrodynamics along the wave flume and over the breakwater can be simulated properly for both solitary wave and tsunami overflow experiments. Stability of the overall cross-section can only be simulated qualitatively for solitary wave cases; on the other hand, the effect of the time elapsed during tsunami overflow cannot be reflected in the simulations using the present numerical tool. However, the stability of the overall cross-section under tsunami overflow is assessed by evaluating forces acting on the rear side armor unit supporting the crown-wall of the rubble mound breakwater as a practical engineering application in the present paper. Furthermore, two non-dimensional parameters are derived to discuss the stability of this armor unit; and thus, the stability condition of the overall cross-section. Approximate threshold values for these non-dimensional parameters are presented comparing experimental and numerical results as a starting point for engineers in practice. Finally, investigations on the solitary wave and tsunami overflow experiments/simulations are extended to the potential tsunami simulation in the scope of both representation of a realistic tsunami in a wave flume and stability of the rubble mound breakwater.

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1. Introduction

After the Great East Japan Tsunami in 2011, small-scale computational fluid dynamics (CFD) simulations became necessary as the tsunami destroyed many bridges, breakwaters, and other coastal infrastructures in an unexpected manner [1]. Although there are various studies focusing assessment of tsunami impact on seawalls,

composite breakwaters, and bridges, there is a limited number of studies investigating the effect of tsunamis on rubble mound breakwaters considering both physical and numerical modeling studies. Esteban et al. [2] studied the stability of armor units of rubble mound breakwaters against tsunami attacks by physical model experiments. Guler et al. [3], the reference paper for the present study, studied the performance of a selected rubble mound breakwater under the attack of tsunamis as a case study. Furthermore, Harbitz et al. [4] tested a similar cross-section to the one used in Guler et al. [3] considering similar wave conditions. On the other hand, Sakakiyama [5] presented both physical and numerical experiments investigating flow fields of tsunamis passing over

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a rubble mound breakwater. In the numerical part of Sakakiyama [5], a numerical model solving Reynolds Averaged Navier-Stokes (RANS) equations and using Volume of Fluid (VOF) method for free-surface tracking [6] used.

In this study, tsunami attack on a rubble mound breakwater is assessed numerically using a Volume-Averaged Reynolds-Averaged Navier-Stokes (VARANS) solver called IHFOAM [7–11] developed in OpenFOAM® environment [12]. Numerical results are compared with the physical model experiments of Guler et al. [3]. The physical model experiments focused on rubble mound breakwater of Haydarpasa Port (Haydarpasa Breakwater, hereafter), located at the southern entrance of Istanbul Bosphorus Strait in the Sea of Marmara, Turkey. Several studies concluded that 35 tsunamis have occurred in the Sea of Marmara [13,14] including the recent 1999 Marmara Event [15–17]. The physical model experiments [3] conducted in 105 m long wave flume of Port and Airport Research Institute (PARI) consist of both solitary wave and tsunami overflow experiments. Solitary wave experiments aimed to understand the effect of the tsunami-like long waves and to find the related tsunami overflow heights. However, Madsen et al. [18] concluded that solitary waves could not adequately represent the effect of tsunamis. Arikawa et al. [19] revealed that stability of breakwaters depends on the duration of the overflow, due to water level difference between the seaside and the harbor side of breakwaters. Therefore, Guler et al. [3] used tsunami overflow experiments in addition to solitary wave experiments to understand the effect of the duration of the overflow during tsunami attack and the tsunamis' action mechanism on the selected coastal structure.

Selected solitary wave and tsunami overflow experiments are simulated for calibration and validation of the numerical solver. Comparisons of numerical and experimental results are performed for water surface elevations and velocity measurements along the wave flume, and time variation of safety factors against sliding and overturning are calculated evaluating stability condition of the crown-wall of the breakwater. In addition to simulations of solitary wave and tsunami overflow experiments, a simulation with a potential tsunami in Haydarpasa Region is presented based on the data of a numerical tsunami assessment study by Aytore et al. [20]. The potential tsunami simulation aims to extend understanding of tsunami attack on a rubble mound breakwater by numerical simulations, as it is not possible to generate a “real” tsunami in a wave flume with the present state of the art technology.

Numerical assessment studies show that hydrodynamics along the wave flume and over the breakwater can be simulated accurately. On the other hand, stability conditions in the solitary wave cases can be qualitatively assessed as the pressure forces around the crown-wall of the breakwater governed the sliding of the crown-wall and failure of the overall cross-section. Furthermore, the effect of the duration of the flow during tsunami overflow cannot be simulated with the present numerical tool since it is not capable of directly evaluating forces acting on the structural units of the rubble mound breakwater; and thus, the motion of them as a function

of time. However, it is still possible to discuss the stability of the overall cross-section under tsunami overflow by evaluating forces acting on the single stone behind the crown-wall at the rear side of the breakwater as presented in the present paper as an engineering application.

This paper is structured as follows: After this brief introduction, an overview of the experiments, and selected experimental cases for numerical simulations are given in Section 2. Next, the numerical model, computational domain, model parameters, and boundary conditions are described in Section 3. Section 4 presents the results of numerical simulations, and Section 5 discusses the possible engineering applications of this type of numerical studies. Finally, conclusions are presented in Section 6.

2. Overview of physical model experiments

Guler et al. [3] used both solitary waves and tsunami overflow tests to represent the effect of tsunamis more relevantly; and thus, to understand the failure mechanism of the breakwater under tsunami attack. In these experiments, solitary waves are generated by piston-type wavemaker, and constant flow over the crown-wall of the breakwater, i.e., tsunami overflow, is achieved by the inflow from a pump at the inlet and outflow by another pump at the outlet. It is seen that tsunami overflow can be more destructive than solitary waves due to the time elapsed during tsunami overflow.

These physical model experiments were conducted in 105 m channel of Port and Airport Research Institute (PARI) in Japan using a Froude type model scale of 1:30. In the experiments, water surface elevations and the flow velocities were measured by eight wave gauges (WGs) and three acoustic Doppler velocimeters (ADV), respectively. Dimensions of the channel, locations of wave gauges and ADVs are shown in Fig. 1, and a closer side view of the rubble mound breakwater is given in Fig. 2 indicating WGs and ADVs on the cross-section.

Three experiments of Guler et al. [3] are selected for calibration and validation of the numerical model. Solitary wave tests with the wave heights of $H=7.5$ cm and $H=10$ cm are selected for calibration and validation of the numerical model, respectively. In the reference experimental study [3] and in the present study, these experiments are named according to the resulting solitary wave heights at WG6 before construction of the cross-section during the wave calibration stage of the physical model experiments. However, we calibrate and validate the numerical model using the measured experimental data after the construction of the cross-section during the actual experimental stage. As the solitary waves are extremely long waves compared to wind waves, the wave heights of solitary waves increase when there is such an obstacle at the end of the cross-section. Therefore, the reader might observe much higher solitary wave heights than the name of the experiments in the figures comparing the physical model experiments and numerical simulations in the further sections. Water surface elevation time-series data without the rubble mound breakwater

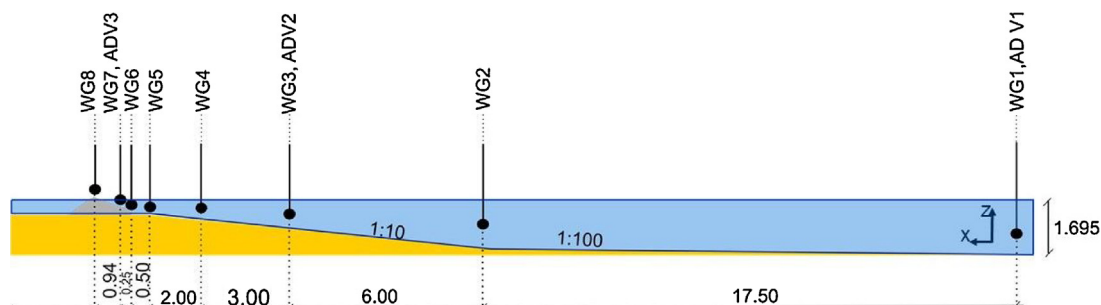


Fig. 1. Cross-Sectional View of Wave Flume (Dimensions are given in meters.).

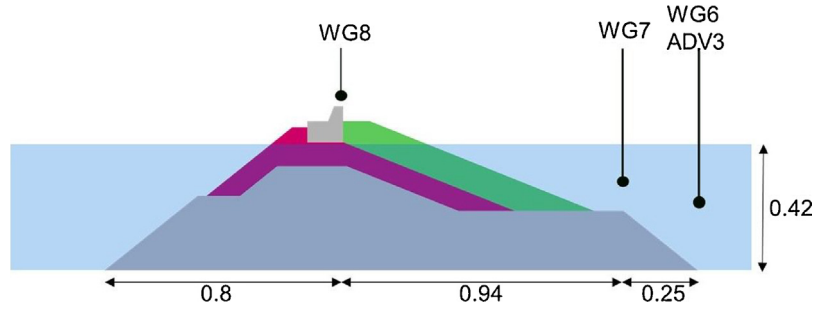


Fig. 2. A Closer Side View of the Rubble Mound Breakwater (Dimensions are given in meters.).

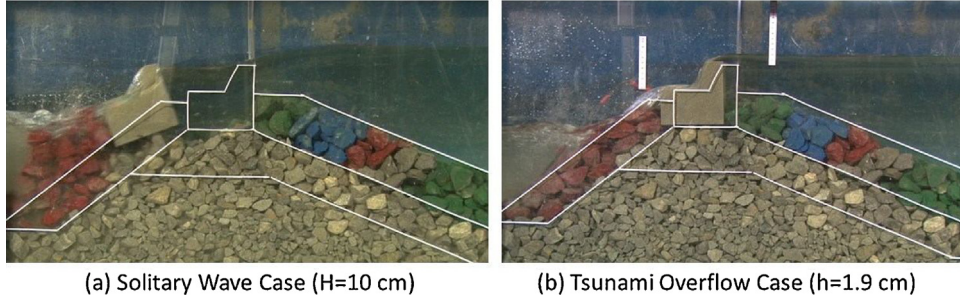


Fig. 3. Snapshots of Haydarpasa Breakwater Crown-Wall's Failures in (a) Solitary Wave Case ($H = 10$ cm), (b) Tsunami Overflow Case ($h = 1.9$ cm) [3].

are also plotted for WG1 to WG7 in the related figures to clarify the solitary wave heights along the wave flume. The first solitary wave test ($H = 7.5$ cm) resulted in minor damage in the cross-section (movement of several stones) whereas the second solitary wave test ($H = 10$ cm) finished with the total failure of the cross-section by sliding of the crown-wall. On the other hand, tsunami overflow test with an overflow height of 1.9 cm is selected to extend the discussions on the elapsed time of flow action which resulted in a total failure by the sliding of the crown-wall. Tsunami overflow height here is defined as the distance between the top of the crown-wall and the water surface elevation above the crown-wall measured by WG8 depicted in Figs. 1 and 2. Furthermore, results of all the tsunami overflow experiments are used for discussions on practical engineering applications. Snapshots from the solitary wave ($H = 10$ cm) and tsunami overflow experiments ($h = 1.9$ cm) are presented in Fig. 3.

3. Description and settings of numerical model

As tsunami attack on a rubble mound breakwater is assessed in the present study, flow inside the porous media must be solved. In a recent extensive review article on this topic presented by Losada et al. [21], approaches to address porous media flow inside porous coastal structures are grouped as macroscopic and microscopic approaches. Macroscopic approaches solve modified flow equations averaging flow properties inside the porous media disregarding the actual geometry (among others [9,10,22,23]). Mean flow properties are captured accurately in macroscopic approach; however, small-scale variations associated with porous media heterogeneities and pore irregularities are smoothed. On the other hand, flow inside the porous media is simulated by constructing actual geometry of the structural units inside the computational domain in microscopic approach. For instance, Dentale et al. [24,25] used microscopic approach to solve flow inside armor units of a rubble mound breakwater. Methodologically, microscopic approach give more accurate results; however, it is rarely used as the mesh generation is relatively challenging, and the computational demand of this method is much higher. In the present study, the macro-

scopic approach is used to solve flow properties inside the porous media. Details of the numerical model and simulations are presented in this section.

3.1. Numerical model: IHFOAM

The numerical model used in the present study is an open-source CFD toolbox, IHFOAM, which was initially developed as the wave generation and absorption boundary conditions in OpenFOAM[®] environment by Higuera et al. [7]. It was successfully applied to several coastal engineering problems in Higuera et al. [8]. Later, Higuera et al. [9] further developed IHFOAM introducing a numerical solver based on the *interFoam* solver of OpenFOAM[®] to solve flow properties in porous media. This new solver was applied to several test cases in Higuera et al. [10]. Recently, a new boundary condition was added to IHFOAM for numerical wave generation with moving boundaries [11].

The *interFoam* solver of OpenFOAM[®] solves Reynolds Averaged Navier-Stokes (RANS) equations in three-dimensions for two incompressible phases, and the free-surface is tracked by the Volume of Fluid (VOF) method given by Rider and Kothe [26]. On the other hand, IHFOAM solves VARANS equations proposed by del Jesus et al. [27] for porous media flow with the free surface. VARANS equations replace RANS equations inside the porous media, and they are equivalent to RANS equations outside the porous media. These equations include conservation of mass (Eq. (1)), conservation of momentum (Eq. (2)), and the VOF advection equation (Eq. (3)).

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (1)$$

$$(1 + c) \frac{\partial}{\partial t} \left(\frac{\rho u_i}{n} \right) + \frac{u_j}{n} \frac{\partial}{\partial x_j} \left(\frac{\rho u_i}{n} \right) = - \frac{\partial p}{\partial x_i} + \rho g_i + \frac{\partial}{\partial x_j} \left(\mu_{eff} \frac{\partial}{\partial x_j} \left(\frac{u_i}{n} \right) \right) - A \frac{u_i}{n} - B \frac{u_i}{n} \left| \frac{u_i}{n} \right| \quad (2)$$

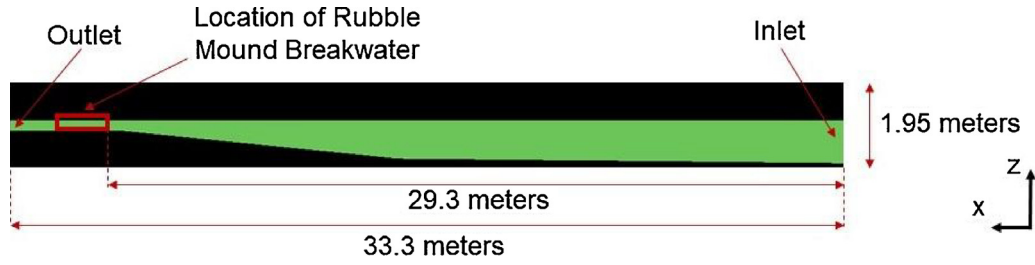


Fig. 4. Side View of Computational Domain (Figure is not to scale.).

$$\frac{\partial \alpha}{\partial t} + \frac{1}{n} \frac{\partial \alpha u_i}{\partial x_i} + \frac{1}{n} \frac{\partial \alpha (1 - \alpha) u_i}{\partial x_i} = 0 \quad (3)$$

In these equations u_i is the extended averaged or Darcy velocity of the i 'th coordinate (x_i); n is the porosity defined as the volume of the voids over the total volume; p is the pressure; ρ is the density; g is the acceleration of gravity; μ_{eff} is the efficient dynamic viscosity defined as $\mu_{eff} = \mu + \rho \nu_{turb}$ where μ is the dynamic molecular viscosity and ν_{turb} is the turbulent kinetic viscosity computed by the selected turbulence model; and α is VOF function. VOF function takes a value of 1 when the cell is full of water; on the other hand, it takes a value of 0 when the cell is empty. A , B and c are the closure parameters related to the physics that cannot be solved when volume-averaging. A , B and c account for frictional forces, pressure forces, and added mass due to the individual elements of the porous media, respectively. Formulas for A and B are respectively given in Eqs. (4) and (5) by van Gent [28] modified from their descriptions presented by Engelund [29].

$$A = a \frac{(1 - n)^3}{n^2} \frac{\mu}{D_{n50}^2} \quad (4)$$

$$B = b \left(1 + \frac{7.5}{KC} \right) \frac{1 - n}{n^2} \frac{\rho}{D_{n50}} \quad (5)$$

where a and b are the friction parameters; D_{n50} is the nominal mean diameter of the material; KC is the Keulegan-Carpenter number introducing additional friction due to the oscillatory nature and unsteadiness of the system:

$$KC = \frac{T_0 u_M}{D_{n50} n} \quad (6)$$

in which T_0 is the period of the oscillation, and u_M is the maximum oscillatory velocity.

Friction parameters a and b have to be calibrated when solving VARANS equations. A thorough literature review is presented in Losada et al. [21] on the values of a and b used in numerical modeling studies of porous media flow. del Jesus et al. [27] has shown that c does not have vital importance in most of the cases, and it is recommended to take its value as 0.34 [9].

OpenFOAM® uses a unique two-step methodology called PIMPLE to solve Navier-Stokes equations. PIMPLE is developed by merging Pressure Implicit Split Operator (PISO) [30] and Semi-Implicit Method of Pressure Linked Equations (SIMPLE) [31] algorithms. The reader is referred to Moukalled et al. [32] for details of the PIMPLE methodology.

Solution of VOF advection equation is carried out by a special solver in OpenFOAM® called Multidimensional Universal Limiter for Explicit Solution (MULES) bounding the result between 0 and 1 using a limiter factor on the fluxes of the discretized divergence term. IHFOAM includes a new version of MULES called IHMULES accounts for the porous media flow [9].

Several turbulence models are already available in OpenFOAM® environment. Higuera et al. [9] adopted volume-averaged $k - \varepsilon$ and

$k - \omega$ SST (shear stress transport) turbulence models to work in IHFOAM based on the derivations given by del Jesus [33]. Thus, these turbulence models work inside and outside the porous media.

3.2. Setting up the numerical model

In this study, two-dimensional (2D) computations are carried out in a numerical tank as the physical model experiments are also 2D. Although the length of the wave flume is 105 m in the experiments, the length of computational domain (horizontal direction, x) is selected as 33.3 m to reduce computational time. The height of the domain (vertical direction, z) is chosen as 1.95 m, and the width (perpendicular direction to the horizontal plane, y) is limited to one cell as 2D simulations are considered. Fig. 4 shows the side view of the computational domain.

Constant mesh sizes of 1 cm along the x -axis and 0.5 cm along z -axis are used in the whole computational domain. Additional refinement is performed around the crown-wall resulted in 0.5 cm mesh size in the x -direction, and 0.25 cm mesh size in the z -direction. This mesh configuration reveals about one million cells. Mesh independency is checked using Grid Convergence Index (GCI) analysis [34] by comparing the parameters computed using finer and coarser mesh configurations. In this analysis, it is usually focused on the differences in parameters such as maximum water surface elevation, maximum velocity, maximum and minimum pressure. In the present study, finer mesh configuration is selected as 0.5 cm along the x -axis and 0.25 cm along the z -axis; furthermore, GCIs are calculated for all WGs and ADVs considering the maximum differences in all the time series obtained from the computations carried out with finer and coarser mesh configurations. Even though, the maximum GCI is found out to be 1.8% indicating mesh independent solution.

Inlet boundary conditions are selected as measured surface elevation at WG1 and measured current velocity at ADV1 for the solitary wave experiments. On the other hand, inlet boundary conditions are forced as water surface elevations and particle velocities reflecting the trend at WG1 and ADV1, respectively, for the tsunami overflow experiments. The measured water surface elevations and particle velocities cannot be inputted as in the case of solitary waves since excessive computational time is required to reach the steady-state condition, i.e., constant overflow. Instead, water surface elevations and particle velocities are increased to the expected level faster than the physical model experiments as the inlet boundary condition. For the final simulation focusing on a potential tsunami in the area of interest, water surface elevations and particle velocities of a sinusoidal wave are used as the inlet boundary condition. Details of the generation of inlet boundary condition are discussed in Section 4.4. Furthermore, the absorbing boundary condition is set at the inlet and the outlet for all simulations. The upper boundary is considered as 'atmosphere' whereas the bottom boundary is set as 'wall'. Finally, type of side walls in the y -direction are selected as "empty". Using "empty type walls" in OpenFOAM® environment along a certain direction converts the

simulation to a 2D computation as the momentum equations are not solved in that direction.

In the present study, $k-\omega$ SST turbulence model developed by Menter [35] is used. $k-\omega$ SST turbulence model is a combination of $k-\varepsilon$ and $k-\omega$ turbulence models as the former model have a better performance in the free flow region, and the latter model has a better performance near the boundaries. The reader is referred to del Jesus [33] and del Jesus et al. [27] for the details of the selected turbulence model.

Simulation duration is selected as 30 s for solitary wave experiments, 25 s for tsunami overflow experiments, and 50 s for potential tsunami case. The computational time on 12 threads (Intel® Xeon® E5-1650 v3 3.5 GHz) with 16GB of RAM is about 24 h, 150 h and 191 h for the solitary wave experiments, tsunami overflow experiments and potential tsunami case, respectively.

4. Results of simulations

4.1. Model calibration (solitary wave case, $H = 7.5$ cm)

The numerical model is calibrated using the data of solitary wave experiment with a solitary wave height of 7.5 cm. This experiment resulted in minor damage. Diameters of stones (D_{n50}) in each layer of rubble mound breakwater, the porosity of the layers and friction parameters are presented in Table 1. Note that friction parameter c is taken as 0.34 for all the layers as discussed earlier in Section 3.

The simulation results show that solitary wave overflows the crown-wall of the rubble mound breakwater between $t = 13.8$ s and $t = 15.2$ s. Snapshots from the simulation indicating the velocity distribution are given in Fig. 5 during the solitary wave attack. In the snapshots, it can be observed that the maximum velocity is around 0.3 m/s, and it is captured at the front of the wave. Velocity magni-

Table 1
Model Parameters.

Layers	Diameter of Stones (D_{n50}) in meters	Porosity (n)	Friction Parameters		
			a	b	c
Armor	0.040	0.40	50	0.6	0.34
Filter	0.033	0.35	50	2.0	0.34
Core	0.015	0.30	50	1.2	0.34

tudes at the rear side are higher than the seaside indicating lower damage at the seaside matching the observations in the physical model experiments. Furthermore, the evolution of flow separation at the seaside toe of the rubble mound breakwater is seen clearly.

The computed time histories of water surface elevations in simulations are compared with the measured time histories in physical model experiments after construction of the breakwater cross-section (with BW) at wave gauges (WG1–WG8) in Fig. 6. In addition, measured time histories before construction of the breakwater (w/o BW) are also plotted in the same figure at wave gauges (WG1–WG7) to compare evolution of the solitary wave with and without the breakwater cross-section. Measured time history without the breakwater cross-section is not plotted at WG8 to stress the tsunami overflow comparison. Fig. 6 shows that the computed water surface elevations are in reasonably good agreement with the physical model experiments with the breakwater cross-section. On the other hand, it is seen from the experimental measurements that reflection from the breakwater cross-section causes difference at all the wave gauges, and significant effect of reflection is observed at wave gauges 5–7. The computed time histories of the horizontal component of particle velocities in the x-direction are also compared with the measured time histories in physical model

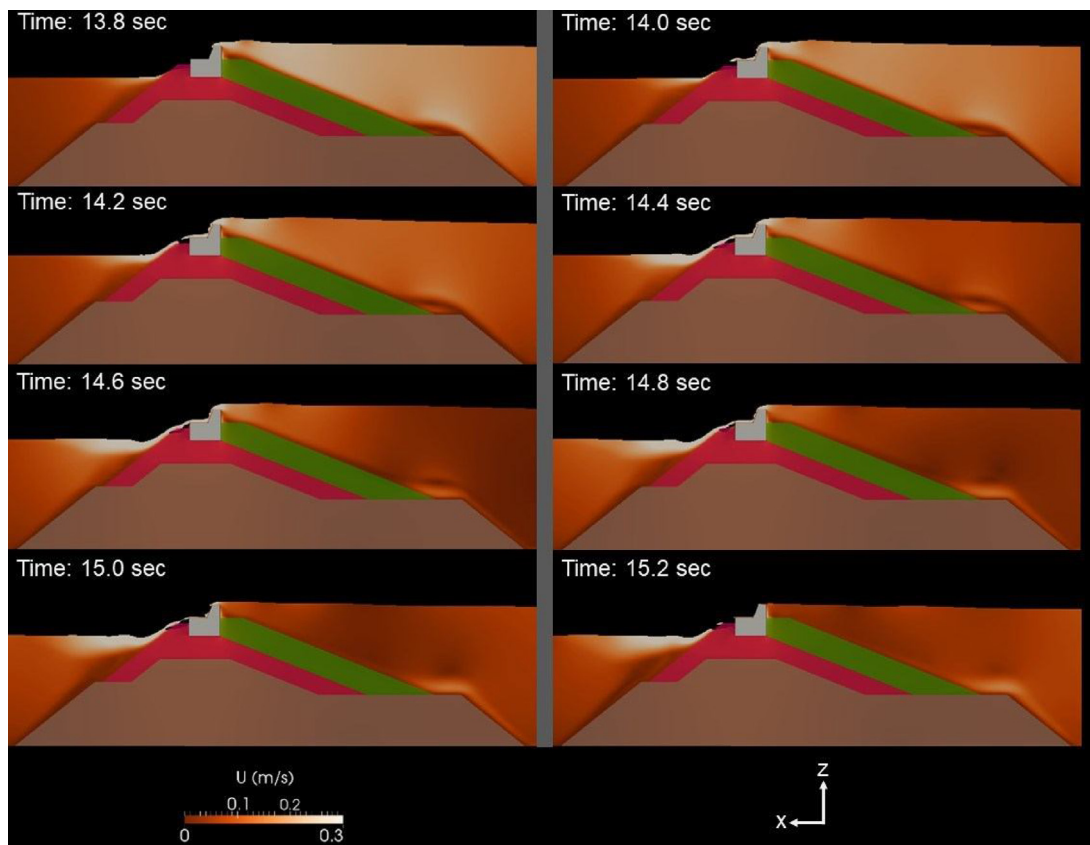


Fig. 5. Snapshots from Simulation of Solitary Wave Case, $H = 7.5$ cm: Velocity Distribution around the Rubble Mound Breakwater during Solitary Wave Attack.

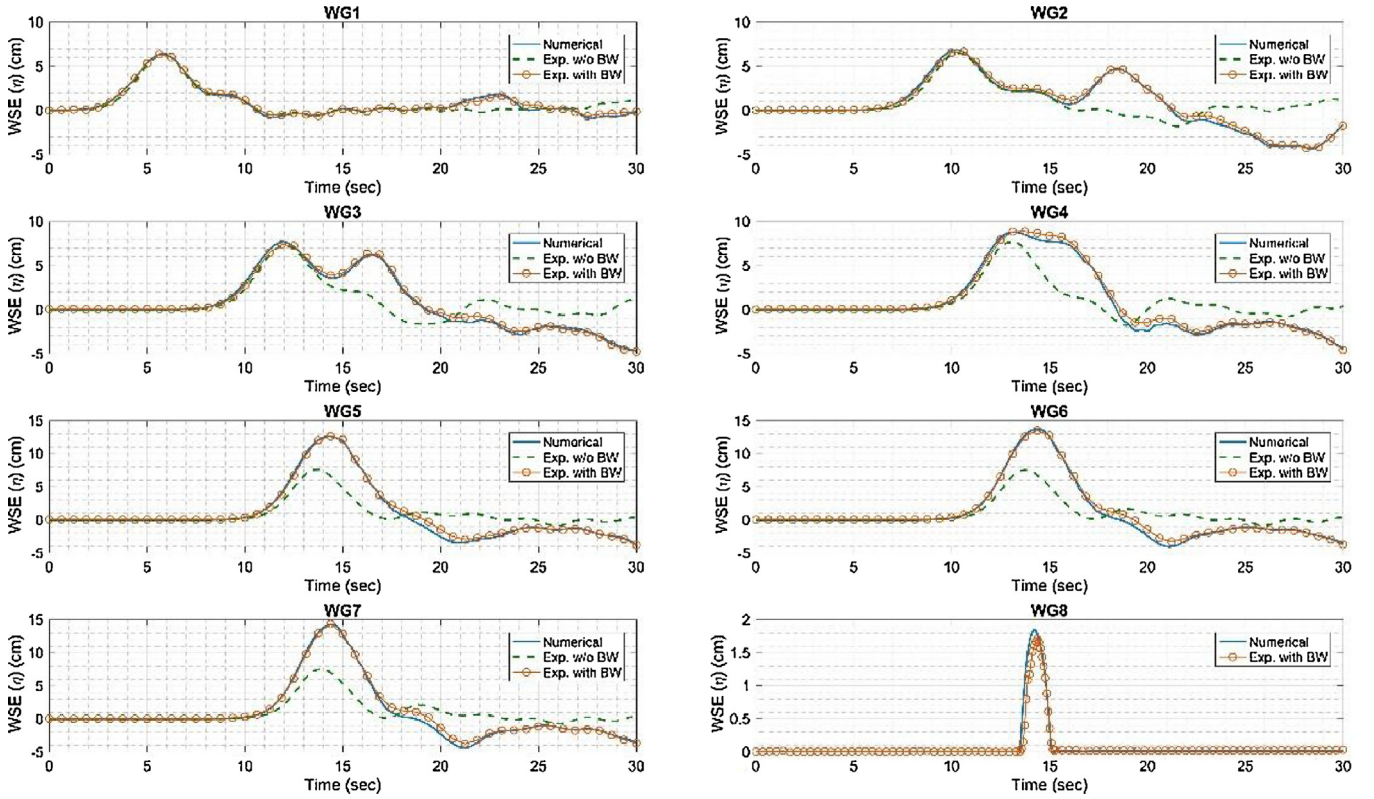


Fig. 6. Comparison of Water Surface Elevations of the Numerical Model (solid blue line) with the Experimental Measurements with the Breakwater Cross-Section (orange line and circles) and without the Breakwater Cross-Section (dashed green line), Solitary Wave Case ($H = 7.5$ cm). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

experiments at ADVs (V1–V3) in Fig. 7. These comparisons show that particle velocities are in good agreement with the physical model experiments with breakwater cross-section. There are some

minor deviations at V3 most probably originated from the volume averaging, or 2D/3D effects after reflection from the breakwater as the ADV is at the toe of the armor layer, or from both. However, it still captures the general trend and the magnitude of the particle velocities.

The variation of safety factors in time is calculated against sliding and overturning by integrating the instantaneous pressure distribution around the crown-wall of the breakwater to discuss the stability of the breakwater. The formulas used for computation of safety factors are given in Eqs. (7) and (8), respectively.

$$SF_{sliding} = \frac{\mu \sum F_z}{\sum F_x} \quad (7)$$

$$SF_{overturning} = \frac{\sum M_{resisting}}{\sum M_{driving}} \quad (8)$$

In Eq. (7), $\sum F_z$ is the vector summation of pressure forces in the vertical (z) direction and the weight of the crown wall, μ is the friction coefficient between the crown-wall and the stones along the crest of the breakwater, and $\sum F_x$ is the vector summation of pressure forces in the horizontal (x) direction. In Eq. (8), $\sum M_{resisting}$ is the summation of resisting moments due to pressure forces and the weight of the crown-wall in the counter-clockwise direction, and $\sum M_{driving}$ is the summation of resisting moments in the clockwise direction. The friction coefficient (μ) and weight of the crown-wall are taken same as in the physical model experiments [3]. In these computations, the support force applied by the stone at the rear side of the breakwater is not taken into account. Contribution from this component is not negligible; however, it is not trivial to estimate supporting forces using the present numerical methodology. This issue is addressed in Section 5.2. as a practical engineering application. Therefore, safety factor calculations should be considered as a qualitative discussion throughout this study. The variation of safety

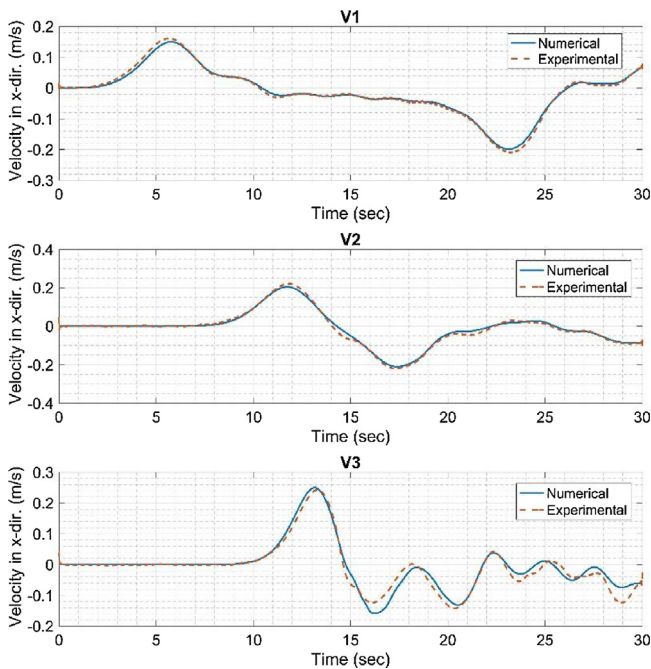


Fig. 7. Comparison of Particle Velocities in x-direction of the Numerical Model (solid blue lines) with the Experimental Measurements (dashed orange lines), Solitary Wave Case ($H = 7.5$ cm). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

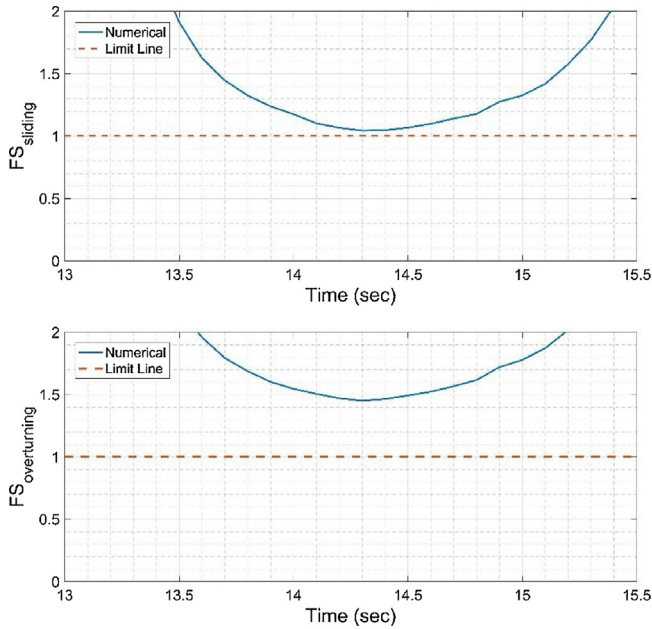


Fig. 8. Time Variation of Safety Factors, Solitary Wave Case ($H = 7.5$ cm).

factors in time is provided in Fig. 8 when the solitary wave is overflowing the crown-wall. It is seen that the crown-wall (from $t = 13$ s to $t = 15.5$ s, where t is the simulation time) is stable ($SF > 1$) against both sliding and overturning as in the physical model experiments.

4.2. Validation (solitary wave case, $H = 10$ cm)

The numerical model is validated against the experimental data of 10 cm solitary wave case. Note that, this experiment resulted in total failure in which failure mechanism is sliding of the crown-wall. Since the time elapsed during the solitary wave attack is rather short compared to overflow experiments, failure is related to the action of mobilizing forces (wave pressure forces) and stabilizing forces (weight, bottom friction force, supporting forces applied to the stones at the rear side). In other words, there is not enough time for significant scouring taking place at the rear side of the breakwater reducing the supporting forces applied by the stones at this side as in the overflow experiments. Therefore, sliding of the crown-wall governs the failure mechanism in the solitary wave experiments.

Similar to the model calibration procedure, the time histories of water surface elevation and particle velocities are compared for validation. The time histories of water surface elevations at gauges WG1–WG8 are presented in Fig. 9 comparing measurements from physical model experiments with breakwater cross-section and numerical results. In addition to that, experimental measurements without breakwater at wave gauges WG1–WG7 are plotted in Fig. 9 to compare evolution of the solitary wave with and without the breakwater cross-section. There is a significant deviation from the experimental data in WG8 located at the top of the crown-wall. As the crown-wall slid in this experiment, water surface elevation suddenly decreases in the experimental data with breakwater cross-section. On the other hand, it is seen from Fig. 9 that reflection from the breakwater cross-section causes difference at all the wave gauges. Significant effect of reflection is observed at the wave gauges 5–7. In Fig. 10, horizontal components of particle velocities at gauges V1–V3 are plotted comparing numerical results with experimental measurements with breakwater cross-section. It is

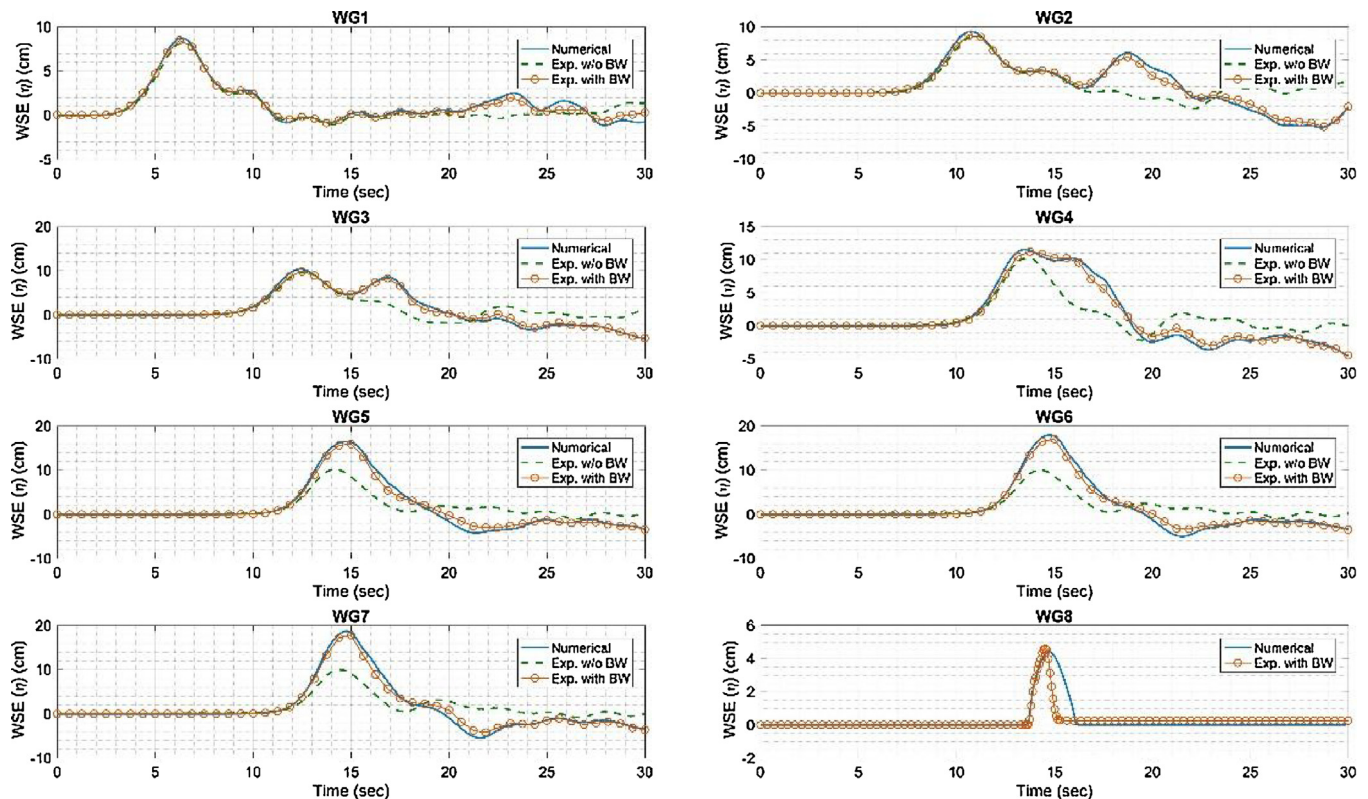


Fig. 9. Comparison of Water Surface Elevations of the Numerical Model (solid blue lines) with the Experimental Measurements (dashed orange lines), Solitary Wave Case ($H = 10$ cm). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

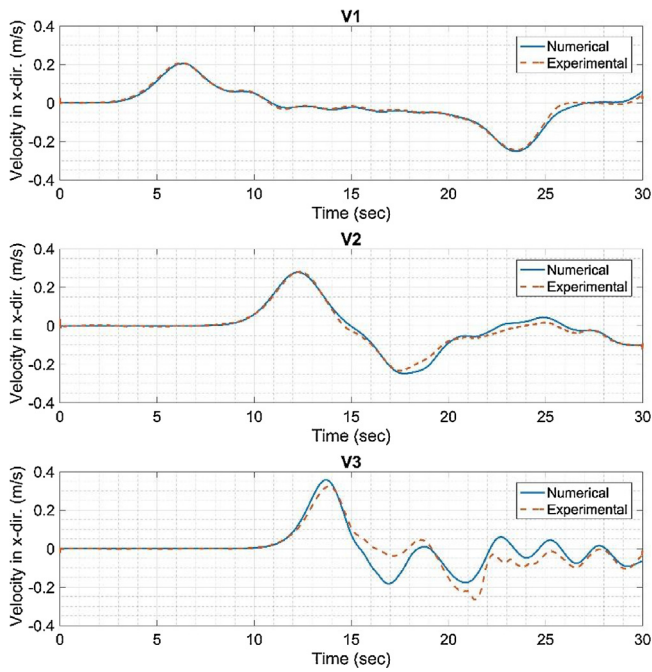


Fig. 10. Comparison of Particle Velocities in x-direction of the Numerical Model (solid blue lines) with the Experimental Measurements (dashed orange lines), Solitary Wave Case ($H = 10$ cm). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

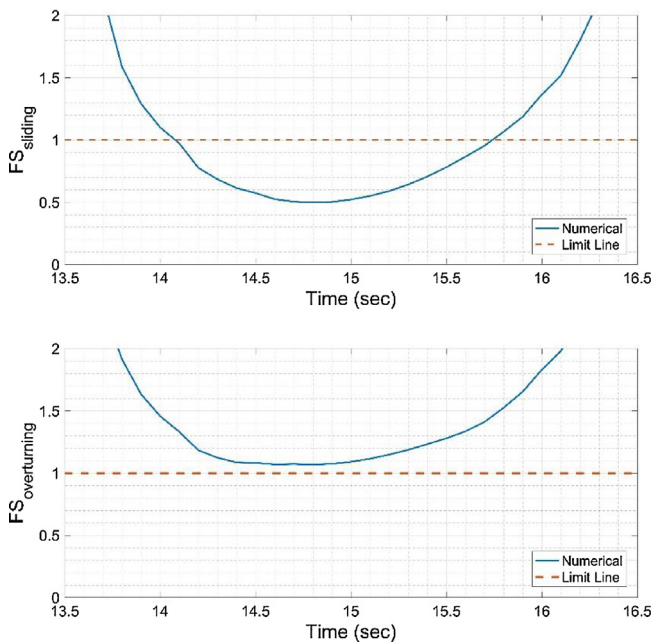


Fig. 11. Time Variation of Safety Factors, Solitary Wave Case ($H = 10$ cm).

seen in Fig. 10 that numerical results match experimental results accurately. The deviation from the experimental result in V3 is observed due to the sliding of the crown-wall around $t = 14.5$ s.

Time variation of safety factors against sliding and overturning are also computed for the validation case by integrating instantaneous pressure around the crown-wall given in Fig. 11. In this case, it is observed from numerical results that crown-wall is not safe ($SF < 1$) against sliding matching with the experimental observation. In the physical model experiment, supporting forces also play a major role in the stabilizing direction; however, non-stabilizing pressure forces govern the process. Therefore, it is reasonable to

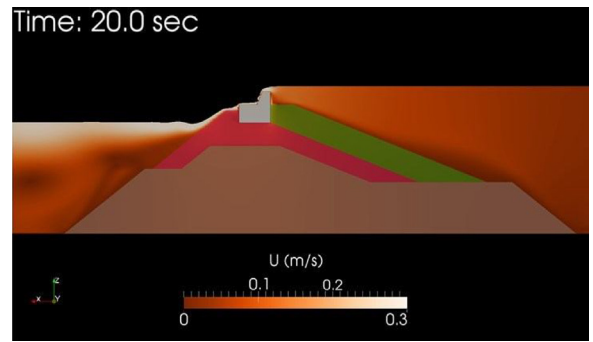


Fig. 12. Snapshot from Simulation of Tsunami Overflow Case at $t = 20$ sec, $h = 1.9$ cm: Velocity Distribution around the Rubble Mound Breakwater during Tsunami Overflow.

say that the numerical model is validated to model time variation in safety factors qualitatively.

4.3. Tsunami overflow case ($h = 1.9$ cm)

In the physical model experiments, tsunami overflow tests are carried out mainly to reflect the effect of the time elapsed during the tsunami overflow. In the physical model experiments, overflow heights of 1.9 cm and 4.6 cm correspond to 7.5 cm and 10 cm solitary wave cases, respectively. In the solitary wave experiments, it is concluded that 7.5 cm solitary wave experiment resulted in minor damage, and the 10 cm solitary wave experiment resulted in total failure. However, in the tsunami overflow experiments, using 1.9 cm tsunami overflow height caused total damage of the cross-section, because the continuous overflow decreased the supporting forces by scouring the stones at the rear side. This contradiction from the physical model experiments indicates the importance of the time elapsed during tsunami overflow.

In the present study, tsunami overflow case ($h = 1.9$ cm) is simulated to validate the model for tsunami overflow cases and to discuss the difference between solitary wave cases and tsunami overflow cases. In this simulation, water surface elevation reaches steady-state condition approximately after 20 s. A snapshot from the simulation is given in Fig. 12 showing the water surface elevation and the velocity distribution around the rubble mound breakwater during tsunami overflow at $t = 20$ s. It is observed from the simulation that the magnitude of particle velocities reach their maximum (0.3 m/s) above the crown-wall. Relatively high particle velocities are seen around the crown-wall and the free-surface at the rear side of the breakwater.

In the physical model experiments, the tsunami overflow is provided by pumps at the inlet side of the wave flume causing high oscillations in the measurements. After the steady-state condition is reached, it is trivial to compare numerical and experimental results. Therefore, results for WG7, WG8, and ADV3 are only presented to show that water surface elevation captured before and at the top of the crown-wall (Fig. 13) noting that the oscillations in the experimental data are related to the surface waves due to the inflow from the pump at the inlet. In Fig. 13, it is seen that there is a significant time lag between experimental and numerical results since water surface elevation and particle velocities are increased to the expected level faster than the physical model experiments as the inlet boundary condition to reduce computational time. Higher water surface elevations and particle velocities are observed just before the steady-state condition is reached due to the same reason. However, the numerical results show that water surface elevations (WG7) and particle velocities (ADV3) at the toe of the breakwater, and overflow height (WG8) are captured at the required level after the steady-state condition is reached.

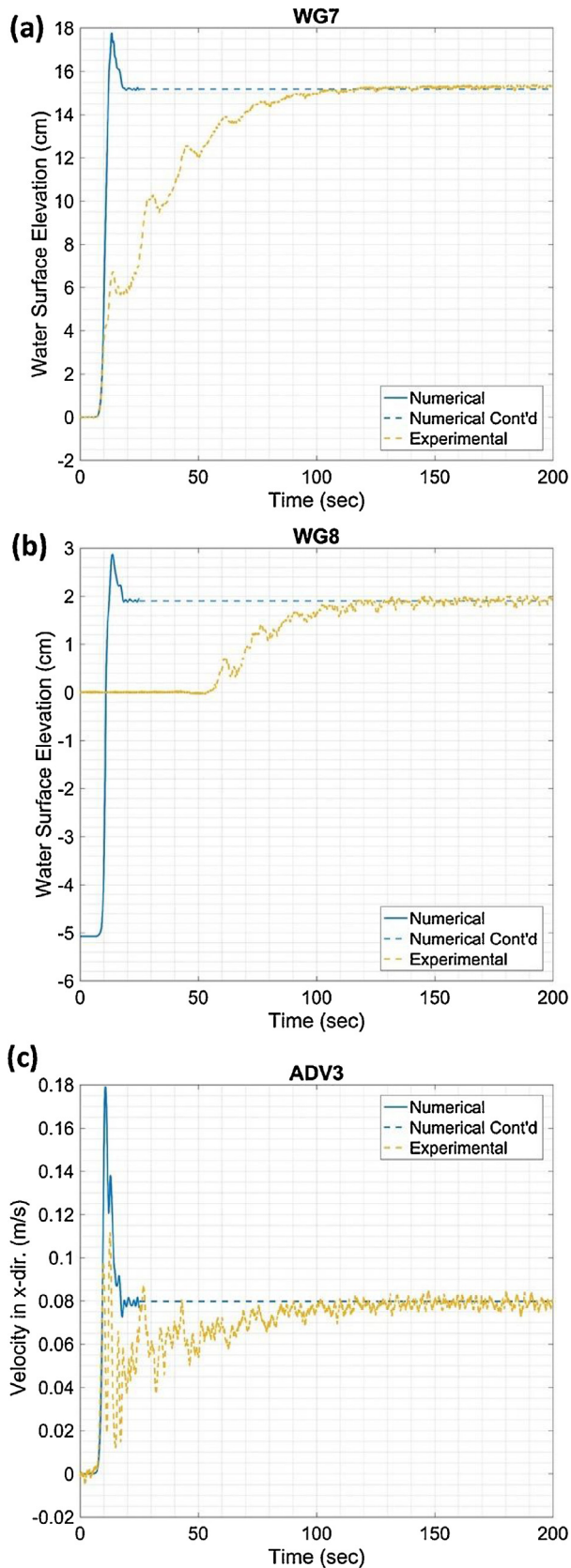


Fig. 13. Comparison of Water Surface Elevation at (a) WG7 and (b) WG8, and Particle Velocities in x-direction at (c) ADV3 of the Numerical Model (numerical results: solid blue lines, continuation of numerical results: dashed blue lines) with the Experimental Measurements (dashed orange lines), Tsunami Overflow Case ($h = 1.9$ cm). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

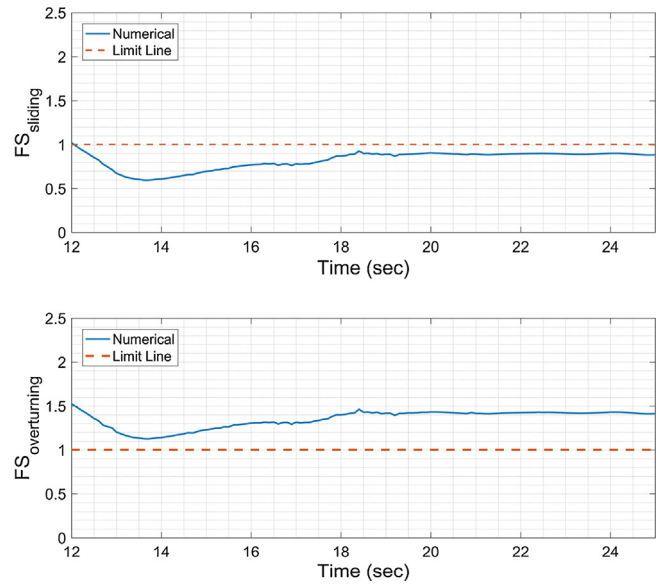


Fig. 14. Time Variation of Safety Factors, Tsunami Overflow Case ($h = 1.9$ cm).

Time variation of the safety factors against sliding and overturning are also computed for the tsunami overflow case presented in Fig. 14. It is seen from the safety factor computations that the crown-wall is not stable against sliding, and is stable against overturning. However, as the factor of safety values against sliding is close to limiting value ($FS = 1$), the contribution from the supporting force by the single stone at the rear side of the breakwater gains more importance than the solitary wave experiments. In other words, the curve showing the variation of factor of safety against sliding might be pushed up due to the supporting force. Therefore, it can be concluded that the present numerical model cannot simulate the failure mechanism in this case properly because the governing mechanism of the failure is sliding of the crown-wall due to the reduction of the supporting force caused by scouring at the rear side due to tsunami overflow which is “the effect of time elapsed during tsunami overflow”. Improvement focusing on the movement of the stones or scouring caused by this type of flows is required. However, it is still possible to discuss the movement of the stones approximately by evaluating forces acting on them under certain assumptions as presented in Section 5.2.

4.4. A potential tsunami case

It is challenging to generate a real tsunami in a wave flume with the present state of the art technology, as a wave generator with a very long stroke length is required. However, it is feasible to generate a more realistic representative tsunami in a numerical simulation since there are no limits on the stroke length. Madsen and Fuhrman [36] showed that sinusoidal waves could represent a real tsunami inspired by observations from the Indian Ocean Tsunami from December 26, 2004, and Larsen et al. [37] recently used this result in a study focused on numerical assessment of tsunami-induced scouring around a monopile foundation. In their paper, Madsen and Fuhrman discuss that although the event is a transient phenomenon, it might be approximated by a regular wave event. They give two example cases from the Indian Ocean Tsunami at the deep water ($h = 2000$ m) and at a relatively shallower water depth ($h = 14$ m). These example cases are represented by the same average wave period ($T = 13$ min) with varying wave amplitudes ($A_0 = 0.75$ m at $h = 2000$ m and $A_0 = 2.5$ m at $h = 14$ m). In the present study, a potential tsunami in Haydarpasa Region is represented by a single sinusoidal wave regarding previous research [36,37].

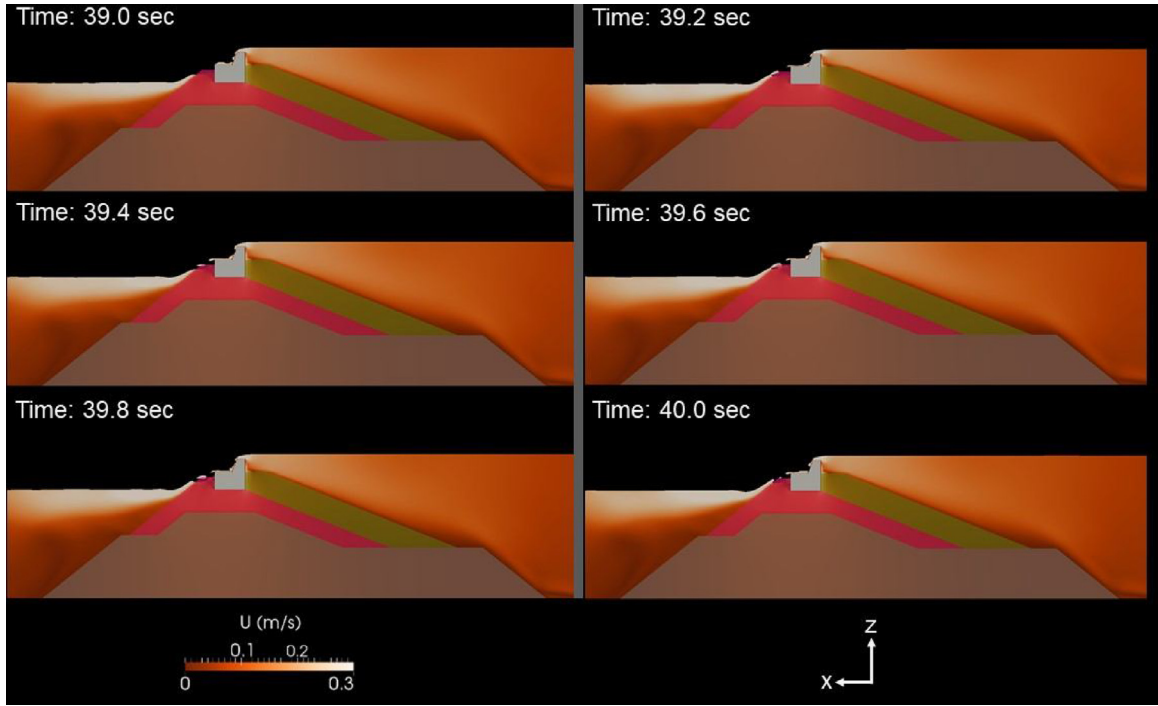


Fig. 15. Snapshots from Simulation of the Potential Tsunami Case: Velocity Distribution around the Rubble Mound Breakwater.

Before describing the selection procedure of the wave amplitude and period of the sinusoidal wave representing the potential tsunami, definitions of wave heights of a solitary wave, a tsunami, and a sinusoidal wave should be clarified as these definitions might differ in different references. In the present study, the solitary wave height and the tsunami height are considered as the distance from the mean water level to the crest of these waves. On the other hand, the distance from mean water level to the crest or the distance from mean water level to the trough of a sinusoidal wave is referred as the wave amplitude as in the most of references. Therefore, the amplitude of the sinusoidal wave is considered as solitary wave height or tsunami height in this study.

The period and amplitude of this wave are selected using the results of a detailed tsunami assessment study with a high-resolution topographic data set in the Sea of Marmara focusing on Haydarpasa Region [20]. A tsunami numerical modeling software called NAMIDANCE [38] is used in this study [20] to simulate tsunami generation, propagation, and inundation based on the solution of non-linear shallow water equations. This study [20] showed that Yalova Normal Fault (YAN) is the most critical source of a tsunami event for Haydarpasa Region. Resulting water surface elevation time-series are presented at eight different locations close to Haydarpasa Region (indicated as cborder3 to cborder11) in Fig. 5 of Aytore et al. [20] for YAN source. Water surface elevation time-series data of Aytore et al. [20] at the eight different locations are investigated by zero-up crossing method to find the typical wave period of a potential tsunami in the region. The average wave period of a potential tsunami in Haydarpasa Region is calculated as 4 min in prototype scale corresponding to 43.8 s in model scale of the reference physical model experiments (a Froude model scale of 1/30). This value is assumed as the representative wave period of all the waves in a potential tsunami event in this region. On the other hand, tsunami heights up to 3.5 m might be expected around Haydarpasa Breakwater for YAN source [20]. In the present study, 2.25 m tsunami height in prototype scale corresponding to 7.5 cm in model scale is selected for the potential tsunami simulation. Thus, the results of the potential tsunami simulation can be compared

with the solitary wave case used for calibration ($H = 7.5$ cm) and the corresponding tsunami overflow case with 1.9 cm overflow height. Solitary wave height for $H = 7.5$ cm case at the WG1 (inlet) is 6.34 cm as presented in Fig. 6. Therefore, the amplitude of the sinusoidal wave representing the potential tsunami is selected as 6.34 cm at the inlet. Note that the water depth at the inlet and in front of the rubble mound breakwater corresponds to 51 m and 12.6 m in prototype scale, respectively. Therefore, the potential tsunami can still be represented by a sinusoidal wave according to discussions of Madsen and Fuhrman [36].

As the wave is assumed to be started with the drawback of water level, water surface elevations and corresponding water particle velocities at the inlet are generated using Eqs. (9) and (10), respectively.

$$\eta(t) = -a \sin(\omega t) \quad (9)$$

$$U(t) = -U_m \sin(\omega t) \quad \text{where } U_m = a\sqrt{\frac{g}{h}} \quad (10)$$

In Eqs. (9) and (10), t represents time, a is the amplitude of the sinusoidal wave taken as 6.34 cm, ω is the angular wave frequency taken as 0.143 s^{-1} , η is water surface elevation, and U is water particle velocity at time t . In Eq. 10, U_m is the velocity magnitude beneath the tsunamis, g is the acceleration of gravity, and h is the water depth at the inlet.

Snapshots from the simulation presenting the velocity distribution around the cross-section of Haydarpasa Breakwater are presented in Fig. 15 between $t = 39.0$ s and $t = 40.0$ s. It is seen that the maximum velocity is approximately 0.3 m/s around the crown-wall and at the free-surface of the rear side of the breakwater.

Time variation of water surface elevations at WG8 representing the tsunami overflow on the top of the crown-wall of both potential tsunami case and the reference solitary wave case ($H = 7.5$ cm) are plotted in Fig. 16. Obviously, the curve for the reference solitary wave case is offset to match the peak water surface elevation value of this case with potential tsunami case. Fig. 16 shows that the tsunami overflow height reaches 1.9 cm for both cases. It is clearly seen from Fig. 16 that the duration of the overflow (7.6 s) is sig-

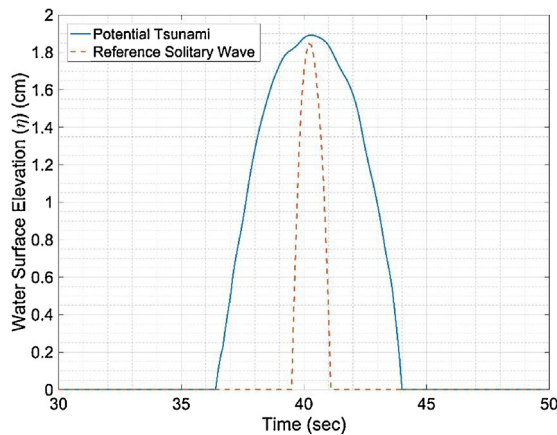


Fig. 16. Water Surface Elevations at WG8 of Potential Tsunami Case (blue line) and Reference Solitary Wave Case ($H = 7.5$ cm) (dashed orange line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

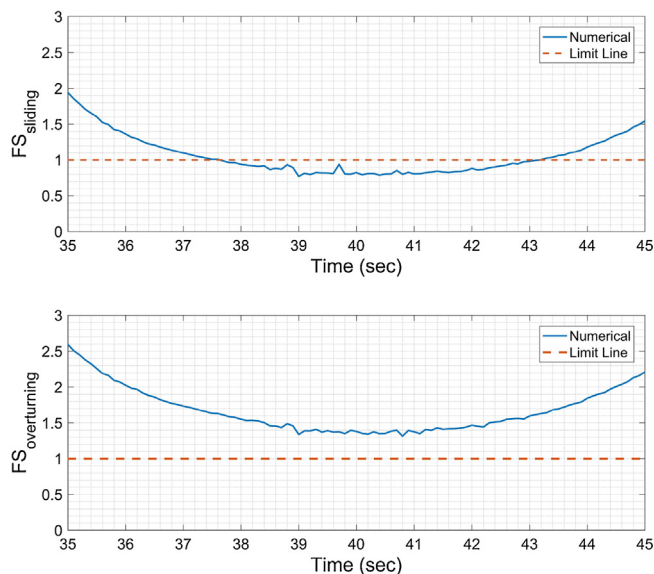


Fig. 17. Time Variation of Safety Factors, Potential Tsunami Case.

nificantly longer in the potential tsunami case than the reference solitary wave case ($H = 7.5$ cm). This result indicates the importance of the tsunami overflow experiments since such a potential tsunami cannot be generated in physical model experiments with the present state of the art technology.

Time variation of safety factors against sliding and overturning are calculated for the potential tsunami case using Eqs. (7) and (8) given in Fig. 17. The crown-wall is found to be not stable against sliding, and stable against overturning. As in the corresponding tsunami overflow case ($h = 1.9$ cm), factor of safety values against sliding are close to the limiting value ($FS = 1$) which indicates that the supporting force due to the stone at the rear side of the breakwater gains more importance. This issue is addressed in Section 5.2.

5. Discussions on practical engineering applications

Advances in the computer technology enable simulation of more and more complicated problems numerically within the shorter computational time. Numerical models make it possible to simulate different type of wave conditions on the different types of coastal structures. Different bathymetric conditions can also be

implemented easily in these simulations. These type of simulations help engineers in practice to compute forces acting on the selected elements of the breakwaters as the velocity and pressure fields around the breakwater are determined in details even for the cases that there are no available analytical or empirical solutions. In this section, two possible applications of these type of simulations to practical engineering applications are presented and discussed.

5.1. Pressure forces around the crown-wall of the breakwater

Although it is possible to estimate pressure distribution at the front edge and the bottom edge of the crown-walls under the attack of wind waves using analytical or empirical relations, it is not trivial to estimate pressure distribution along the other edges of the crown-walls. Furthermore, there is not an explicit formula for pressure distribution around the crown-wall of a rubble mound breakwater under the attack of solitary waves or tsunami overflow to the associated literature. However, it is possible to obtain pressure distributions all around the crown-wall of a rubble mound breakwater at any time step as in the present paper.

In Fig. 18 the evolution of the pressure distribution around the crown-wall when 7.5 cm solitary wave is attacking to the breakwater between time $t = 14.2$ s and $t = 14.8$ s are presented. In the figure, crown-wall dimensions are normalized by the solitary wave height (H), and pressure distributions around the crown-wall are normalized by dividing pressure values with $\rho g H$ where ρ is the density of the water, and g is the acceleration of gravity. It is seen from the results that maximum pressure is applied to crown-wall from the bottom edge, i.e., uplift pressure, during solitary wave action. It is observed that the magnitude of the primary pressure forces (uplift pressure and the pressure force at the seaside edge of the crown-wall) do not change significantly during the wave action; however, change in the magnitude and the direction of the pressure forces can be seen along the other edges of the crown-wall. Notably, it might be useful to observe the change in magnitudes and directions for engineers in practice along the edges for which explicit formulas do not exist. Moreover, one might calculate stability against sliding and overturning using the information related to wave pressures around the crown-wall and Eqs. (7) and (8) as presented before in Section 4.

5.2. Evaluation of forces acting on the single stone behind the crown-wall at the rear side of the breakwater

In Section 4, it is concluded that the present numerical tool is not capable of determining the stability condition of the crown-wall of the rubble mound breakwater as movement or scouring of the stones at the rear side of the breakwater cannot be simulated. However, approximate computations can be carried out to evaluate forces acting on the single stone behind the crown-wall at the rear side of the breakwater – hereafter single stone – under certain assumptions simplifying the problem. Thus, it would be possible to determine the stability condition for both the single stone and the crown-wall of the breakwater.

In the literature, several researchers discussed the forces acting on the armor units of rubble mound breakwaters. Kobayashi and Greenwald [39] and Kobayashi and Otta [40] evaluated drag, inertia and lift forces along a slope with the use of a numerical model solving shallow-water equations. On the other hand, Melby and Hughes [41] and Moghim and Alizadeh [42] focused analyses of physical model experiments based on a parameter called wave momentum flux. Indeed, the authors of the wave momentum flux approach considered parallel and normal forces also corresponding to the drag, inertia and lift forces acting on the armor units. Recent approaches couple flow solvers with discrete element method solvers to simulate damage in the armor layer of the rubble mound breakwaters

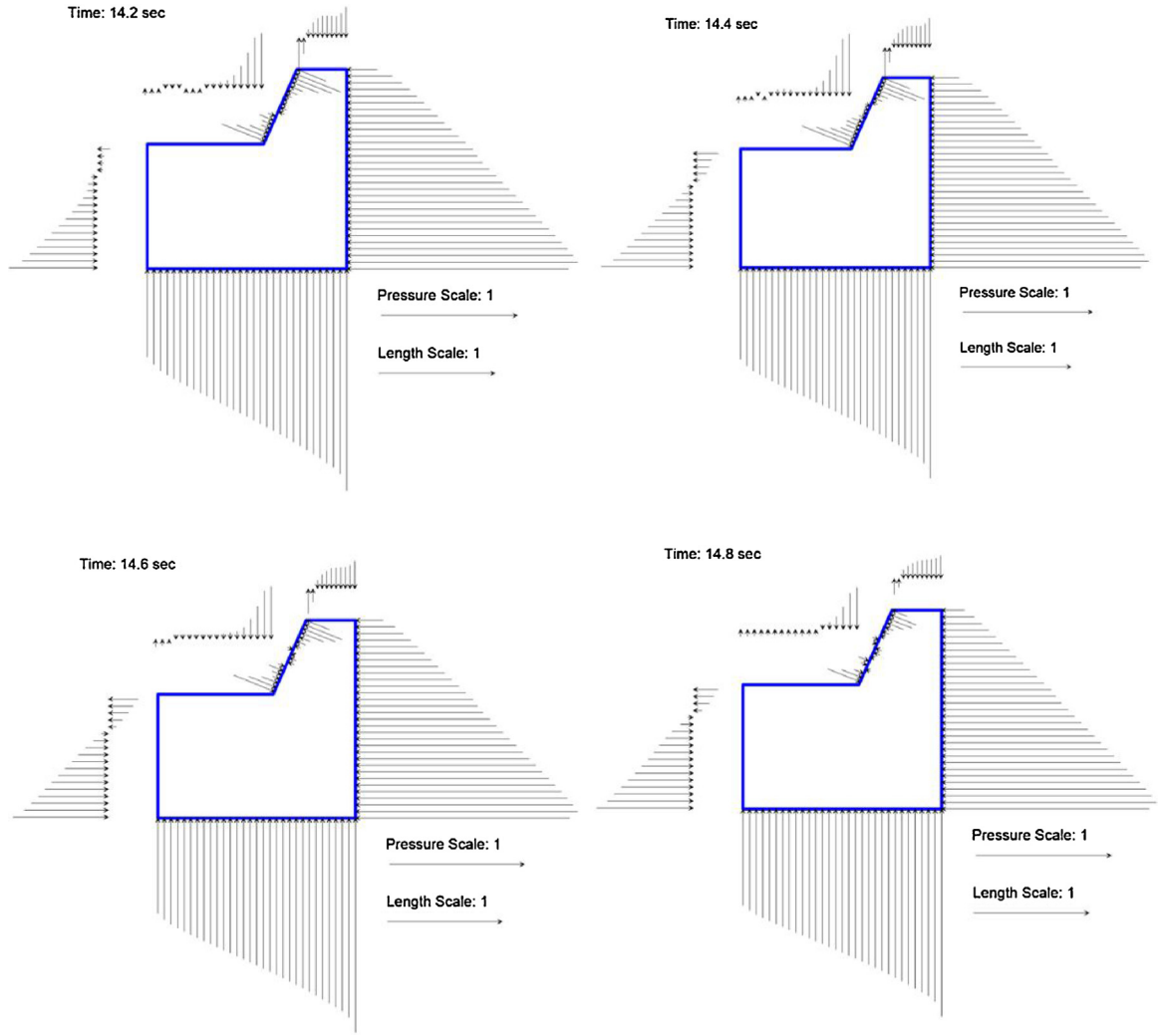


Fig. 18. Evolution of the Pressure Distribution around the Crown-Wall, Solitary Wave Case ($H = 7.5$ cm).

[43,44]. In these approaches, the actual geometry is constructed in the computational domain similar to the studies of Dentale et al. [24,25].

In the present study, all the forces acting on the single stone are derived theoretically and evaluated for several simulations to discuss the stability condition by comparing the numerical results with experimental results. The major assumption in these discussions is that the crown-wall slides when the single stone moves, i.e., slides or rotates, causing total failure of the overall cross-section.

In Fig. 19 a free-body diagram for the single stone is given. In this free-body diagram, F_D and F_I are drag and inertia forces, respectively, acting in the flow direction 1 shown in Fig. 19. Furthermore, F_L is the lift force, W_{sub} is the submerged weight of the stone, and F_H is the horizontal transferred force due to the pressure field around the crown-wall in the flow direction 2 shown in Fig. 19. In the free-body diagram, k stands for the distance from the center of gravity of the single stone to the contact point that the single stone would probably roll about, i.e., k is the moment arm.

Drag force (F_D), inertia force (F_I) and lift force (F_L) are given by Eqs. (11)–(13) respectively.

$$F_D = \frac{1}{2} C_D \rho U_{ave}^2 \left(\frac{\pi D_{50}^2}{4} \right) \quad (11)$$

$$F_I = \frac{1}{2} C_M \rho \frac{dU_{ave}}{dt} \left(\frac{\pi D_{50}^2}{4} \right) \quad (12)$$

$$F_L = \frac{1}{2} C_L \rho U_{ave}^2 \left(\frac{\pi D_{50}^2}{4} \right) \quad (13)$$

where ρ is the density of the water which is 1000 kg/m^3 , U_{ave} is the average flow velocity, and D_{50} is the nominal diameter of the single stone. In these equations, C_D , C_M , and C_L are the drag, inertia and lift coefficients, respectively, and du/dt stands for the instantaneous acceleration computed using average velocity. U_{ave} is calculated at each time step during the flow over the single stone by integrating the velocity profile along the line perpendicular to the flow direction 1. The existence of water particles along the perpendicular line, i.e., the cells containing water particles along the perpendicular line, is determined by checking the value of VOF function with a limiting value of 0.5. Flow velocities along this line are computed by solving VARANS equations; however, as there is no porous medium along this line, the equations are practically RANS equations.

The definition for the horizontal transferred force (F_H) due to the pressure field around the crown-wall is given in Eq. (14).

$$F_H = \sum F_x - f_s = \sum F_x - \mu \sum F_z \quad (14)$$

where $\sum F_x$ is the vector summation of pressure forces in the horizontal (x) direction, f_s is the friction force between the crown-wall

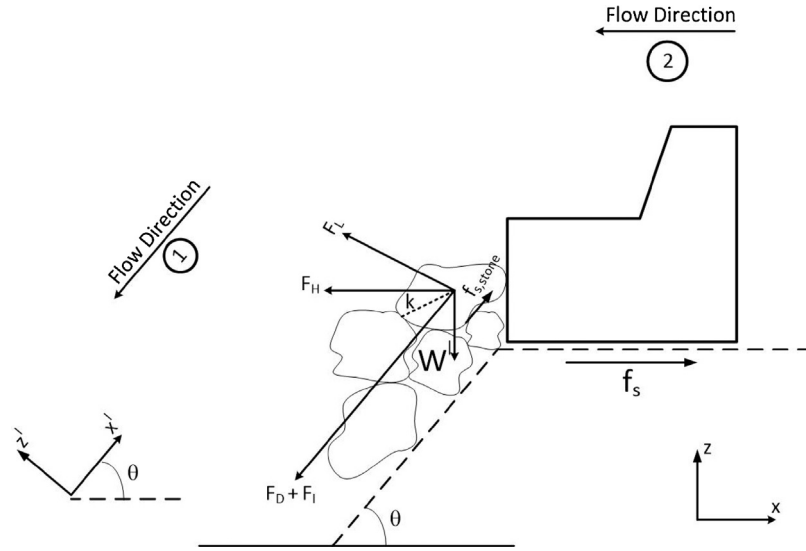


Fig. 19. Free-Body Diagram of Forces Acting on the Single Stone.

and underlying stones, $\sum F_z$ is the vector summation of pressure forces in the vertical (z) direction and the weight of the crown wall, μ is the friction coefficient between the crown-wall and the stones along the crest of the breakwater.

Submerged weight of the stone is computed as

$$W_{sub} = (\rho_s - \rho) D_{50}^3 \quad (15)$$

where ρ_s is the density of the stones which is 2650 kg/m³.

to second stability condition is not observed in the physical model experiments.

After writing the forces acting on the single stone based on the free-body diagram given in Fig. 19 first (Eq. (16)) and third (Eq. (19)) stability conditions are re-written by collecting the terms that can be calculated using the results of numerical simulations in the stability equations at the right-hand side as presented in Eqs. (20) and (21), respectively.

$$C_D + C_L \tan \phi \leq \frac{W_{sub} \cos \theta \tan \phi - F_H \sin \theta \tan \phi - F_H \cos \theta - W_{sub} \sin \theta - C_M \rho \frac{dU_{ave}}{dt} \left(\frac{\pi D_{50}^2}{4} \right)}{\frac{1}{2} \rho U_{ave}^2 \left(\frac{\pi D_{50}^2}{4} \right)} \quad (20)$$

$$C_D \cos \phi + C_L \sin \phi \leq \frac{W_{sub} \sin (\phi - \theta) - F_H (\sin \theta \sin \phi + \cos \theta \cos \phi) - C_M \rho \frac{dU_{ave}}{dt} \left(\frac{\pi D_{50}^2}{4} \right) \cos \phi}{\frac{1}{2} \rho U_{ave}^2 \left(\frac{\pi D_{50}^2}{4} \right)} \quad (21)$$

The first stability condition is written along x^l direction (sliding along x^l direction) given by Eq. (16).

$$F_D + F_I + F_H \cos \theta + W_{sub} \sin \theta \leq f_{s, stone} \quad (16)$$

where θ is the structure slope angle which is 39°, and $f_{s, stone}$ is the friction force between the single stone and the underlying layer of stones. $f_{s, thestone}$ is given by Eq. (17) as

$$f_{s, stone} = (W_{sub} \cos \theta - F_L - F_H \sin \theta) \tan \phi \quad (17)$$

where ϕ is the angle of repose.

The second stability condition is written along z^l direction (movement normal to the structure slope) given by Eq. (18).

$$F_L + F_H \sin \theta \leq W_{sub} \cos \theta \quad (18)$$

The third and the final stability condition is written for rolling about the contact point as given in Eq. (19).

$$k [(F_L + F_H \sin \theta) \sin \phi + (F_H \cos \theta + F_D + F_I) \cos \phi] \leq k W_{sub} \sin (\phi - \theta) \quad (19)$$

In writing these stability conditions, stone angularity or shape, surface friction, and stone interlocking are not included. Furthermore, only the first (Eq. (16)) and third (Eq. (19)) stability conditions are taken into account in further computations as the failure due

Right-hand sides of Eqs. (18) and (19) are two non-dimensional parameters denoted as S and R , respectively, where S stands for sliding based on the first stability condition (Eq. (16)) and R stands for rolling based on the third stability condition (Eq. (19)). These parameters are used to relate the stability condition mathematically to non-dimensional parameters. Calculations of these parameters are carried out using inertia coefficient (C_M) as 1.5 and angle of repose (ϕ) as 50° [39,40] for several cases at each time step when there is flow over the single stone. By comparing the numerical results with the experimental data, it is aimed to find critical threshold values for S and R that defines the motion of the single stone; thus, the stability condition of the overall cross-section. The computations are first carried out for tsunami overflow experiment with an overflow height (h) of 1.9 cm. Time variation of non-dimensional S and R parameters are presented in Fig. 20 with respect to time.

In addition to the presented tsunami overflow simulation ($h = 1.9$ cm), tsunami overflow simulations are extended with seven more simulations having tsunami overflow heights ranging between $h = 1.05$ cm and $h = 2.15$ cm to increase the amount of the numerical data for comparisons as Guler et al. [3] presented experimental results for tsunami overflow heights between $h = 1.1$ cm and $h = 1.95$ cm. It is observed from Fig. 20 that there are oscillations in both non-dimensional parameters mainly due to high oscillations in the flow velocity over the single stone before the flow reaches a steady-state condition. Therefore, averages of these

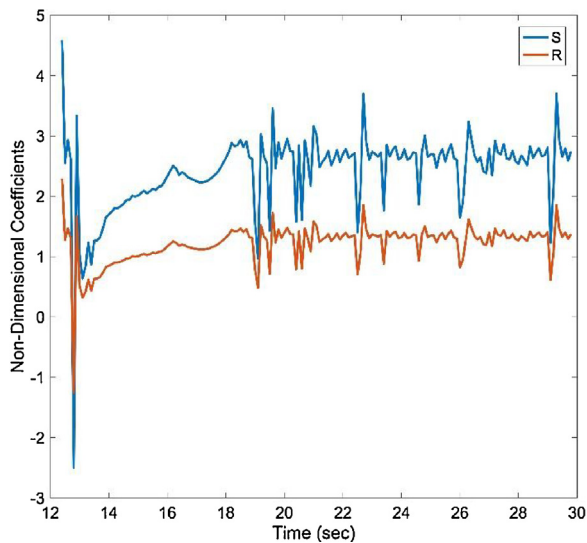


Fig. 20. Time Variation of the Non-Dimensional Coefficients ($\bar{S}$ and $\bar{R}$) for Tsunami Overflow Case ($h = 1.9$ cm).

parameters ($\bar{S}$ and $\bar{R}$) are taken between 20 and 30 s to represent the non-dimensional parameters for each simulation, and the both the experimental and numerical results are presented in Table 2. In $\bar{S}$ and $\bar{R}$ computations, the main uncertainty is related to the inertia coefficient (C_M) as it should be determined empirically. However, as the acceleration over the single stone is not significant, inertia force corresponds to 5–7% of the driving forces meaning that the uncertainty due to this force can be considered as acceptable for these approximate calculations.

Table 2 shows that $\bar{S}$ and $\bar{R}$ decrease as the tsunami overflow height increases. This result indicates that lower $\bar{S}$ and $\bar{R}$ values correspond to decrease in stability. Furthermore, it is seen from both Table 2 and Fig. 20 that rolling about the contact point is more critical than sliding along the slope of the structure.

The non-dimensional $\bar{S}$ and $\bar{R}$ parameters are also computed for solitary wave experiment with a solitary wave height of $H = 7.5$ cm. As the overflow time is short (approximately 1 s) for solitary wave experiments, the average of S and R parameters are taken for a time interval of 0.5 s around the peak overflow height due to the solitary wave attack is observed assuming 50% of the overflow time is effective for the motion of the single stone. $\bar{S}$ and $\bar{R}$ are found as 2.98 and 1.45, respectively. These results are also in the no damage region comparing the numerical results presented in Table 2. Other solitary wave experiment ($H = 10$ cm) is not considered as $\bar{S}$ and $\bar{R}$ take negative values since F_H dominates the computations. The non-dimensional $\bar{S}$ and $\bar{R}$ parameters are also calculated for potential tsunami case. The average of S and R parameters are taken for a time interval of 3.8 s around the peak overflow height due to

the potential tsunami based on the same assumption for solitary wave cases, i.e., assuming 50% of the overflow time is effective for the motion of the single stone. $\bar{S}$ and $\bar{R}$ are computed as 2.49 and 1.25, respectively. These values indicate the failure of the cross-section in regard to Table 2. Note that this result coincides with the corresponding tsunami overflow case ($h = 1.9$ cm).

The experimental results given in Table 2 reveal that there is a gray region between $h = 1.7$ cm and $h = 1.9$ cm as the stability condition changes between failure and no damage conditions. Comparing the experimental results with numerical results and staying on the safe side for the practical engineering applications, one may conclude that $\bar{S}$ bigger than 2.8 and $\bar{R}$ larger than 1.45 indicate stability of the single stone, and thus, the crown-wall and the overall cross-section. In other words, an engineer in practice might interpret the non-dimensional $\bar{S}$ and $\bar{R}$ parameters computed using results of tsunami overflow simulations for a similar conventional rubble mound breakwater with the threshold values mentioned above for these parameters. At this point, it should be noted that these results strongly depend on structure geometry, flow conditions, numerical model parameters, and grid resolution. However, considering these dependencies, one may use these threshold values as a starting point.

6. Conclusions

In the present study, tsunami attack on a rubble mound breakwater is assessed numerically using a solver called IHFOAM developed in OpenFOAM® environment. Data from several physical model experiments [3] are used for calibration, validation, and further discussions. Furthermore, a potential tsunami simulation is utilized to extend discussions related to the representation of a tsunami in physical and numerical tests.

Comparisons of water surface elevations and particle velocities show that numerical results are in fairly well agreement with the experimental data for the solitary wave experiments. Furthermore, the time variations of safety factors against sliding and overturning calculated by integrating instantaneous pressure forces around the crown-wall of the breakwater indicate that failure mechanism can be simulated qualitatively using the present numerical tool for solitary wave experiments.

Numerical modeling of tsunami overflow experiments concludes that hydrodynamics around the rubble mound breakwater can be simulated accurately. However, it is not possible to model the effect of time elapsed until failure during the tsunami overflow using the present numerical tool. It is required to solve or reflect interlocking forces between stones to solve numerically the effect of time elapsed during tsunami overflow.

Results of the numerical modeling studies are discussed in the scope of the practical engineering applications. It is shown that it is possible to obtain the evolution of pressure distribution around the crown-wall of a rubble mound breakwater. Furthermore, forces acting on the single stone located behind the crown-wall at the

Table 2
Comparison of Experimental and Numerical Results.

Physical Model Experiment #	h (cm)	Stability Condition	Numerical Test #	h (cm)	$\bar{S}$	$\bar{R}$
1	1.1	No damage	tr25	1.05	3.29	1.65
2	1.15	No damage	tr26	1.22	3.13	1.56
3	1.2	No damage	tr27	1.4	2.98	1.45
4	1.3	No damage	tr28	1.63	2.87	1.39
5	1.4	No damage	tr29	1.75	2.66	1.34
6	1.7	Failure	tr30	1.9	2.62	1.31
7	1.85	No damage	tr31	2.0	2.58	1.29
8	1.9	Failure	tr32	2.15	2.35	1.17
9	1.95	Failure				

rear side of the breakwater are derived theoretically, three stability conditions are written, and two non-dimensional parameters are presented under certain assumptions based on these forces for sliding of the single stone and rolling of the single stone along the slope of the cross-section. It is seen that the rolling of the single stone is more critical in the context of stability. Approximate threshold values related to the motion of this single stone, and thus, the stability condition of the overall cross-section, are determined by comparing experimental and numerical results. An engineer in practice might use these approximate threshold values as a starting point for the utilization of the results of similar simulations. The assumptions and the uncertainties stated previously in the present paper must be considered carefully in such an application.

Results of the simulation with the potential tsunami show the importance of tsunami overflow experiments since the crown-wall is stable in the corresponding solitary wave experiment ($H = 7.5$ cm) and the crown-wall is not stable in the corresponding tsunami overflow experiment ($h = 1.9$ cm). In other words, the result from the reference physical model experiments [3] indicating the importance of the time elapsed during tsunami overflow is demonstrated by the present numerical study.

Overall, this study concludes that the present numerical tool is capable of simulating tsunami attack on a rubble mound breakwater in the scope of hydrodynamics. However, even if failure mode can be simulated qualitatively for solitary wave experiments and motion of stones can be discussed based on the numerical results, it is still required to improve the numerical tool by including the effect of movement of stones or scouring in the rubble layers.

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